TE – 105 ENGINEERING THERMODYNAMICS
UNIT 1V / PART A

1. Define an air standard cycle. Why are such cycles conceived? (NOV 2017)

If we use air as working substance in the thermodynamic cycles, it is called air standard cycles.

To simplify the analysis of I.C engines, air standard cycles are conceived.

In an air standard cycle, a certain mass of air operates in a complete thermodynamic cycle, where heat is added and rejected with external heat reservoirs, and all the processes in the cycle are reversible.

2. How do you express the efficiency of a mixed or dual cycle? (NOV 2017)

Efficiency of dual cycle

$$\eta_{dual} = 1 - \frac{1}{r^{\gamma-1}}\left[\frac{r_p(\rho^{\gamma} - 1)}{(r_p - 1) + \gamma r_p(\rho - 1)}\right]$$

r = compression ratio
r_p = pressure ratio
ρ = Cut off ratio
γ = adiabatic Index

3. Define Mean effective pressure.(MAY 2019, NOV 2016, JAN 2016)

Mean effective pressure is defined as the constant pressure acting on the piston during the working stroke. It is also defined as the ratio of work done to the stroke volume of piston displacement volume.

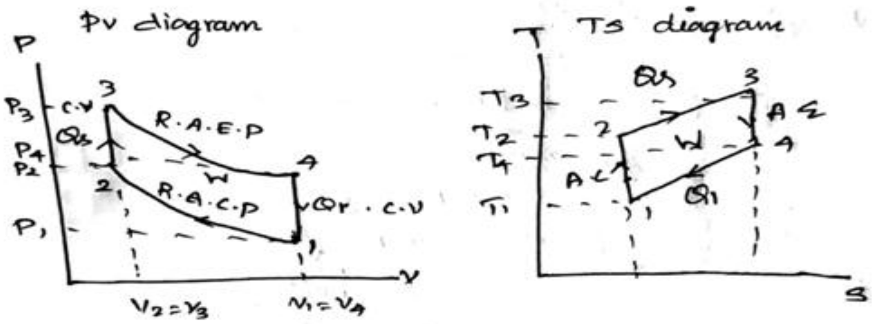
4. How does the change in compression ratio affect the air standard efficiency of an otto cycle? (MAY 2019)

The efficiency of the air standard Otto cycle is thus a function of the compression ratio only. The higher the compression ratio, the higher the efficiency it is independent of the temperature levels at which the cycle operates. The compression ratio cannot, however be increased beyond a certain limit, because of the noisy and destructive combustion phenomenon, know as detonation.

5. List the assumptions for air standard cycles. (MAY 2016, JAN 2016)

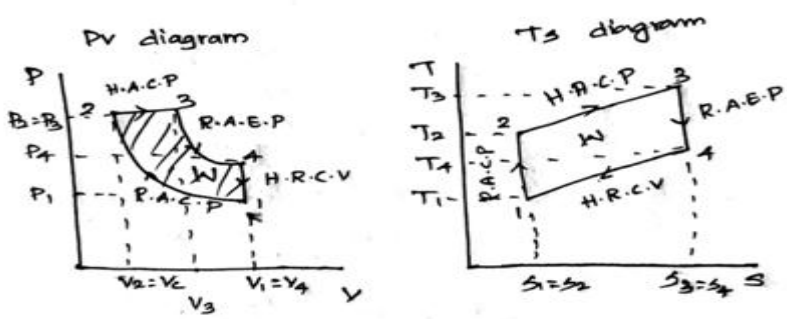
- The working medium is prefect gas throughout i.e. it follows PV = mRT
- The working medium has constant specific heats
- The working medium does not undergo any chemical change throughout the cycle
- The compression and expansion processes are reversible adiabatic i.e. there is no loss or gain in entropy.

6. Draw P-V and T-S diagram for Otto cycle. (JAN 2015)



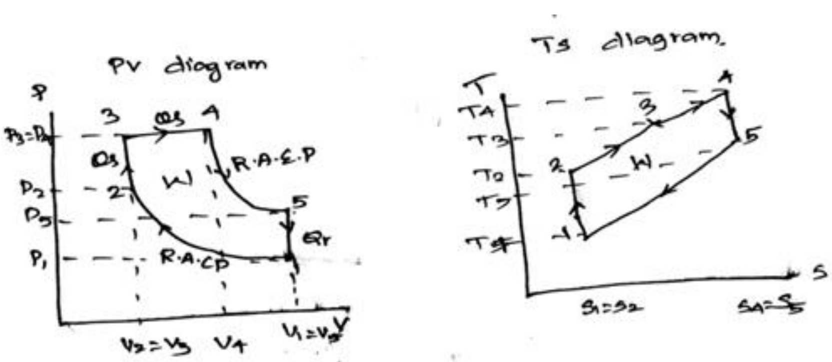
1-2 Reversible adiabatic or isentropic compression process
2-3 Heat addition at constant volume process
3-4 Reversible adiabatic or isentropic expansion process
4-1 Heat rejection at constant volume process

7. Draw P-V diagram for limited pressure cycle or diesel cycle. (JAN 2015, MAY 2018)



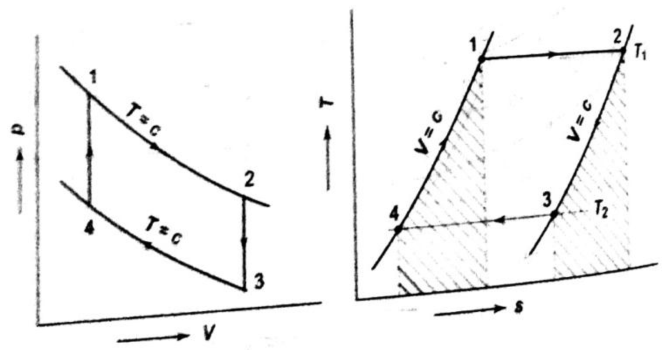
1-2 Reversible adiabatic or isentropic compression process
2-3 Heat addition at constant pressure process
3-4 Reversible adiabatic or isentropic expansion process
4-1 Heat rejection at constant volume process

8. Draw P-v and T-s diagram for dual cycle. (MAY 2016, SEP 2020)



1-2 Reversible adiabatic or isentropic compression process
2-3 Heat addition at constant volume process
3-4 Heat addition at constant pressure process
4-5 Reversible adiabatic or isentropic expansion process
5-1 Heat rejection at constant volume process

9. What are the four processes which constitute the Stirling cycle? (NOV 2018)



1-2 Constant temperature or isothermal process
2-3 Constant volume or isochoric process
3-4 Constant temperature or isothermal process
4-5 Constant volume or isochoric process

10. What is the application of the closed cycle gas turbine plant (NOV 2018)

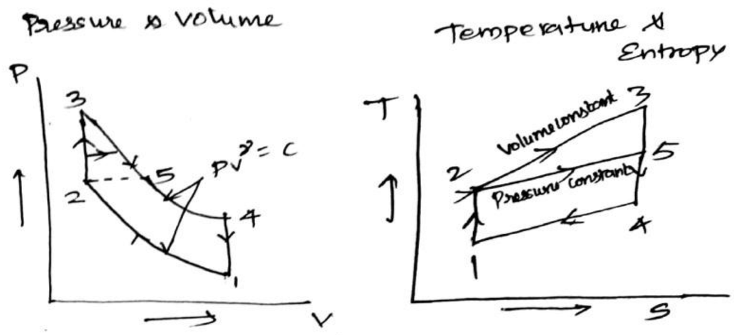
The jet engine is widely used in aviation because they are relatively lightweight and are capable of huge power output.

It allows higher efficiencies at lower temperatures and has a better part load behavior. They allow an operation at higher pressure and the used gases have better thermodynamic properties

11. Define the term: Brayton Cycle. (MAY 2017)

The Brayton cycle is the air standard cycle for the gas turbine power plant. Here air is first compressed reversibly and adiabatically, heat is added to it reversibly at constant pressure, air expands in the turbine reversibly and adiabatically, and heat is then rejected from the air reversibly at constant pressure to bring it to the initial state. The Brayton cycle, therefore consists of two reversible isobars and two reversible adiabatic.

12. For same compression ratio efficiency of Otto cycle is greater than diesel cycle. Why? (MAY 2018)



the two cycles can be compared on the basis of Either the same compression ratio radio or the same maximum pressure temperature

the above PV & Ts diagram shows the compression of otto and diesel cycles for the same compression ratio and heat rejection here

1-2-3-4 otto cycle ; 1-2-5-4 diesel cycle.

For the same Q_2, the higher the Q_1, the higher is the cycle efficiency. In the T-s diagram the area under 2-3 represents Q_1 for the otto cycle, the area under 2-5 represents Q_1 for the diesel cycle. There fore, For the same r_k & Q_2

$\eta_{otto} > \eta_{diesel}$.

13. How does the ideal diesel cycle differ from the ideal Otto cycle?(SEP 2020)

- Diesel cycle is differing from Otto cycle only in heat addition. In Otto cycle, it's a constant volume process whereas in Diesel cycle it's a constant pressure process.
- Compression ratio of Diesel cycle is higher than Otto cycle

14. Define the volumetric efficiency of the air compressor. (MAY 2017)

The ratio of the actual volume of gas taken into the cylinder during suction stroke to the piston displacement volume or the swept volume of the piston is called the volumetric efficiency

$$\eta_{vol} = mv_1/v_s$$

η_{vol} = volumetric efficiency
m = mass of the gas in kg
v_1 = actual volume in m^3
v_s = stroke or displacement volume in m^3

1. **An oil Engine working on theoretical diesel cycle as a bore of 200 mm and stroke of 300 mm. compression is 16. Cutoff takes place at 10% of stroke. Find clearance volume, Cutoff ratio, expansion ratio and air standard efficiency. (JAN 2015)**

given data

$d = 200\,mm = 0.2\,m$

$L = 300\,mm = 0.3\,m$

$r = \frac{V_1}{V_2} = 16$

Cutoff = 10% Stroke : $V_3 - V_2 = 10\% (V_1 - V_2)$

Diesel cycle.

To find

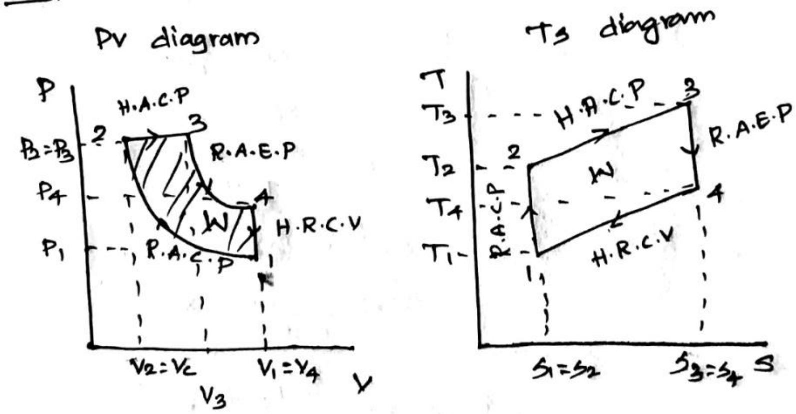
$V_c = V_2$ = clearance volume.

Cut off ratio = $\rho = V_3/V_2$

Expansion ratio = $r/\rho = V_4/V_3$

η_{diesel}

Sol



1) Stroke volume. $V_s = \pi/4\, d^2 \times L$

$$= \pi/4 \times (0.2)^2 \times 0.3$$

$$V_s = 0.0094\ m^3$$

$$r = \frac{V_1}{V_2} = \frac{V_c + V_s}{V_c}$$

$$16 = \frac{V_c + 0.0094}{V_c}$$

$$16 = 1 + \frac{0.0094}{V_c}$$

$$16 - 1 = \frac{0.0094}{V_c}$$

$$15 = \frac{0.0094}{V_c}$$

$$\boxed{V_c = 0.000626 \ m^3}$$

2) Cut off ratio $= \rho = V_3/V_2$

Cut off takes place 10% of stroke

$$V_3 - V_2 = 10\% \ V_1 - V_2 \qquad \frac{V_1}{V_2} = 16 = r$$

$$V_3 - V_2 = 0.1 \ V_1 - V_2. \qquad V_1 = 16 V_2$$

$$V_3 - V_2 = 0.1 \left[16 V_2 - V_2\right]$$

$$V_3 - V_2 = 0.1 [15 V_2]$$

$$V_3 - V_2 = 1.5 V_2$$

$$V_3 = 1.5 V_2 + V_2$$

$$V_3 = 2.5 V_2$$

$$\boxed{\rho = \frac{V_3}{V_2} = 2.5}$$

3) Expansion ratio. $r/\rho = V_4/V_3$

$$r/\rho = V_4/V_3 = 16/2.5$$

$$\boxed{V_4/V_3 = P = 6.4}$$

4) $$\eta_{diesel} = 1 - \frac{1}{\gamma (r)^{\gamma-1}} \left[\frac{\rho^{\gamma} - 1}{\rho - 1}\right]$$

$$= 1 - \frac{1}{1.4 (16)^{1.4-1}} \left[\frac{2.5^{1.4} - 1}{2.5 - 1}\right]$$

$$\boxed{\eta_{diesel} = 0.59 \quad 59\%.}$$

2. **An Engine working on otto cycle has cylinder diameter 150 mm and stroke 225 mm. the clearance volume is 1.25 x 10^{-3} m^3. Find air standard efficiency assuming γ = 1.4. What is the temperature at the end of the compression if initial temperature 40°C. (JAN 2015)**

given data

$d = 150 mm = 0.15 m$

$L = 225 mm = 0.225 m$

$V_c = V_2 = 1.25 \times 10^{-3} m^3$

$\gamma = 1.4$

otto cycle

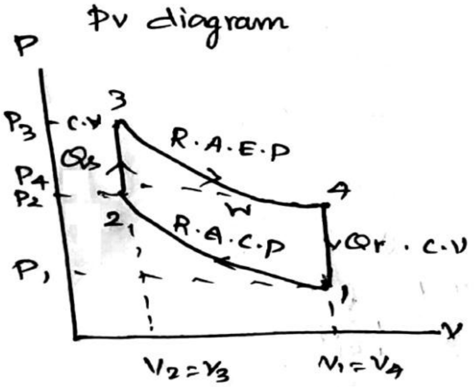
$T_1 = 40°c = 313 K$

To find

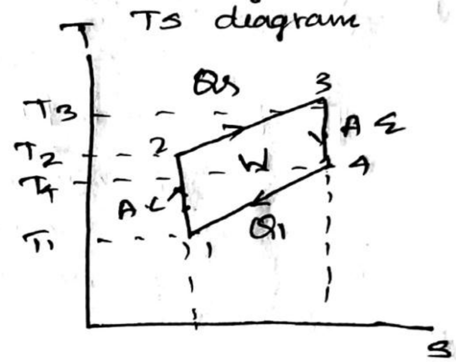
T_2,

__Sol__

pv diagram



Ts diagram



Compression ratio $r = \frac{V_1}{V_2} = \frac{V_c + V_s}{V_c}$

$r = \frac{\text{total cylinder volume}}{\text{clearance volume.}}$

$V_s = \pi/4 \, d^2 \times L$

$= \pi/4 \times (0.15)^2 \times 0.225$

$\underline{\underline{V_s = 0.0039 m^3}}$

$r = \frac{V_c + V_s}{V_c}$

$= \frac{1.25 \times 10^{-3} + 0.0039}{1.25 \times 10^{-3}}$

$$r = \frac{V_1}{V_2} = 4.12$$

$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$$

$$= 1 - \frac{1}{4.12^{1.4-1}}$$

$$\boxed{\eta_{otto} = 0.43 = 43\%}$$

1-2 Reversible Adiabatic process $PV^{\gamma} = C$

~~Press~~ Temperature vs Volume relation

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\frac{313}{T_2} = \left(\frac{1}{4.12}\right)^{1.4-1}$$

$$\boxed{T_2 = 551.44\,K}$$

3. **Drive the expression for an air standard efficiency and mean effective pressure of a diesel cycle (JAN 2016)**

This cycle was introduced by Dr. R. Diesel in 1897. It differs from Otto cycle in that *heat is supplied at constant pressure instead of at constant volume*. Fig. 13.15 (*a* and *b*) shows the *p-v* and *T-s* diagrams of this cycle respectively.

This cycle comprises of the following ***operations*** :

(*i*) 1-2.....*Adiabatic compression.*

(*ii*) 2-3.....*Addition of heat at constant pressure.*

(*iii*) 3-4.....*Adiabatic expansion.*

(*iv*) 4-1.....*Rejection of heat at constant volume.*

Point 1 represents that the cylinder is full of air. Let p_1, V_1 and T_1 be the corresponding pressure, volume and absolute temperature. The piston then compresses the air adiabatically (*i.e.*, pV^γ = constant) till the values become p_2, V_2 and T_2 respectively (at the end of the stroke) at point 2. Heat is then added from a hot body at a constant pressure. During this addition of heat let

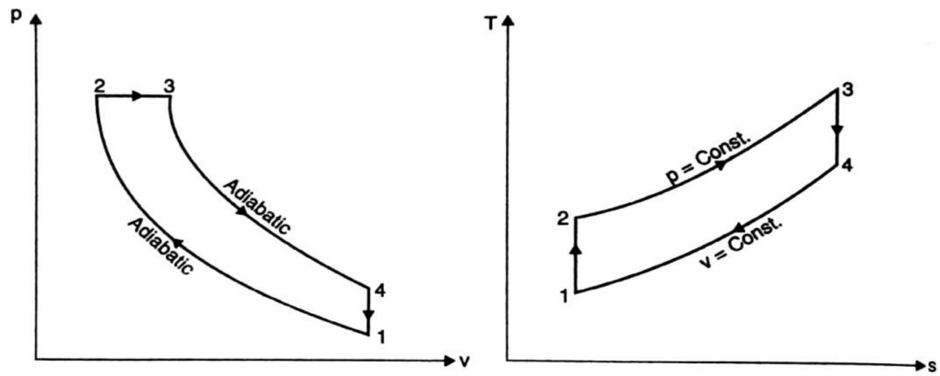


Fig. 13.15

volume increases from V_2 to V_3 and temperature T_2 to T_3, corresponding to point 3. This point (3) is called the ***point of cut-off***. The air then expands adiabatically to the conditions p_4, V_4 and T_4 respectively corresponding to point 4. Finally, the air rejects the heat to the cold body at constant volume till the point 1 where it returns to its original state.

Consider 1 kg of air.

Heat supplied at constant pressure $= c_p(T_3 - T_2)$

Heat rejected at constant volume $= c_v(T_4 - T_1)$

Work done = Heat supplied – heat rejected

$$= c_p(T_3 - T_2) - c_v(T_4 - T_1)$$

$$\therefore \qquad \eta_{diesel} = \frac{\text{Work done}}{\text{Heat supplied}}$$

$$= \frac{c_p(T_3 - T_2) - c_v(T_4 - T_1)}{c_p(T_3 - T_2)}$$

$$= 1 - \frac{(T_4 - T_1)}{\gamma(T_3 - T_2)} \qquad ...(i) \left[\because \frac{c_p}{c_v} = \gamma\right]$$

Let compression ratio, $r = \dfrac{v_1}{v_2}$, and cut-off ratio, $\rho = \dfrac{v_3}{v_2}$ *i.e.*, $\dfrac{\text{Volume at cut-off}}{\text{Clearance volume}}$

Now, during *adiabatic compression 1-2*,

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma - 1} = (r)^{\gamma - 1} \quad \text{or} \quad T_2 = T_1 \,.\, (r)^{\gamma - 1}$$

During *constant pressure process 2-3*,

$$\frac{T_3}{T_2} = \frac{v_3}{v_2} = \rho \quad \text{or} \quad T_3 = \rho \,.\, T_2 = \rho \,.\, T_1 \,.\, (r)^{\gamma - 1}$$

During *adiabatic expansion 3-4*

$$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma - 1}$$

$$= \left(\frac{r}{\rho}\right)^{\gamma-1} \qquad \left(\because \ \frac{v_4}{v_3} = \frac{v_1}{v_3} = \frac{v_1}{v_2} \times \frac{v_2}{v_3} = \frac{r}{\rho}\right)$$

$$\therefore \qquad T_4 = \frac{T_3}{\left(\frac{r}{\rho}\right)^{\gamma-1}} = \frac{\rho \, . \, T_1 (r)^{\gamma-1}}{\left(\frac{r}{\rho}\right)^{\gamma-1}} = T_1 \, . \, \rho^{\gamma}$$

By inserting values of T_2, T_3 and T_4 in eqn. (*i*), we get

$$\eta_{\text{diesel}} = 1 - \frac{(T_1 \, . \, \rho^{\gamma} - T_1)}{\gamma \, [\rho \, . \, T_1 \, . \, (r)^{\gamma-1} - T_1 \, . \, (r)^{\gamma-1}]} = 1 - \frac{(\rho^{\gamma} - 1)}{\gamma (r)^{\gamma-1} (\rho - 1)}$$

or

$$\eta_{\text{diesel}} = 1 - \frac{1}{\gamma (r)^{\gamma-1}} \left[\frac{\rho^{\gamma} - 1}{\rho - 1}\right] \qquad \text{...(13.7)}$$

It may be observed that eqn. (13.7) for efficiency of diesel cycle is different from that of the Otto cycle only in bracketed factor. This factor is always greater than unity, because $\rho > 1$. Hence *for a given compression ratio, the Otto cycle is more efficient*.

The *net work* for diesel cycle can be expressed in terms of pv as follows :

$$W = p_2(v_3 - v_2) + \frac{p_3 v_3 - p_4 v_4}{\gamma - 1} - \frac{p_2 v_2 - p_1 v_1}{\gamma - 1}$$

$$= p_2 \, (\rho v_2 - v_2) + \frac{p_3 \rho v_2 - p_4 r v_2}{\gamma - 1} - \frac{p_2 v_2 - p_1 r v_2}{\gamma - 1}$$

$$\left[\begin{array}{l} \because \ \dfrac{v_3}{v_2} = \rho \quad \therefore \ v_3 = \rho v_2 \ \text{ and } \dfrac{v_1}{v_2} = r \quad \therefore \ v_1 = r v_2 \\ \text{But } v_4 = v_1 \quad \therefore \quad v_4 = r v_2 \end{array}\right]$$

$$= p_2 v_2 \, (\rho - 1) + \frac{p_3 \, \rho v_2 - p_4 r v_2}{\gamma - 1} - \frac{p_2 v_2 - p_1 r v_2}{\gamma - 1}$$

$$= \frac{v_2 \, [p_2 \, (\rho - 1)(\gamma - 1) + p_3 \rho - p_4 r - (p_2 - p_1 r)]}{\gamma - 1}$$

$$= \frac{v_2 \left[p_2 \, (\rho - 1)(\gamma - 1) + p_3 \left(\rho - \dfrac{p_4 r}{p_3}\right) - p_2 \left(1 - \dfrac{p_1 r}{p_2}\right)\right]}{\gamma - 1}$$

$$= \frac{p_2 v_2 \, [(\rho - 1)(\gamma - 1) + \rho - \rho^{\gamma} \, . \, r^{1-\gamma} - (1 - r^{1-\gamma})]}{\gamma - 1}$$

$$\left[\because \ \frac{p_4}{p_3} = \left(\frac{v_3}{v_4}\right)^{\gamma} = \left(\frac{\rho}{r}\right)^{\gamma} = \rho^{\gamma} r^{-\gamma}\right]$$

$$= \frac{p_1 v_1 r^{\gamma-1} \, [(\rho - 1)(\gamma - 1) + \rho - \rho^{\gamma} r^{1-\gamma} - (1 - r^{1-\gamma})}{\gamma - 1}$$

$$\left[\because \ \frac{p_2}{p_1} = \left(\frac{v_1}{v_2}\right)^{\gamma} \ \text{ or } \ p_2 = p_1 \, . \, r^{\gamma} \text{ and } \frac{v_1}{v_2} = r \ \text{ or } \ v_2 = v_1 \, . \, r^{-1}\right]$$

$$= \frac{p_1 v_1 r^{\gamma-1} \, [\gamma(\rho - 1) - r^{1-\gamma} \, (\rho^{\gamma} - 1)]}{(\gamma - 1)} \qquad \text{...(13.8)}$$

Mean effective pressure $\mathbf{p_m}$ is given by :

$$p_m = \frac{p_1 v_1 r^{\gamma-1} \, [\gamma(\rho - 1) - r^{1-\gamma} \, (\rho^{\gamma} - 1)]}{(\gamma - 1) v_1 \left(\dfrac{r - 1}{r}\right)}$$

$$\mathbf{p_m} = \frac{p_1 r^{\gamma} \, [\gamma(\rho - 1) - r^{1-\gamma} \, (\rho^{\gamma} - 1)]}{(\gamma - 1)(r - 1)} \, .$$

4. A gas turbine works on an air-standard Braytoncycle. The initial condition of the air is 20°C and 1 bar. The maximum pressure and temperature are limited to 3 bar and 850°C. Determine the following.

a. Cycle efficiency.
b. Heat supplied and heat rejected per kg of air.
c. Work output per kg of air.
d. Exhaust temperature. (MAY 2019)

given:

$T_1 = 20°C = 293 K$

$P_1 = P_4 = 1 bar \Rightarrow 100 kN/m^2$

$P_2 = P_3 = 3 bar \Rightarrow 300 kN/m^2$

$T_3 = 850°C = 1123 K$

To find:

Brayton cycle efficiency (η_B), Heat supplied (Q_s), Heat rejected (Q_r), Workdone (W), Exhaust temperature (T_4).

Solution:

$$T_2 = T_1 \times \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$= 293 \times \left(\frac{300}{100}\right)^{\frac{1.4-1}{1.4}}$$

$$T_2 = 401.369 K$$

$$\frac{T_4}{T_3} = \left(\frac{P_4}{P_3}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\Rightarrow T_4 = T_3 \times \left(\frac{1}{r_p}\right)^{\frac{1.4-1}{1.4}}$$

$$= 1124 \times \left(\frac{1}{3}\right)^{\frac{1.4-1}{1.4}}$$

$$\boxed{T_4 = 820.906 K}$$

$$Q_s = m C_p (T_3 - T_2) = 1.005 (1123 - 401.369)$$

$$\boxed{Q_s = 725.239 kJ/kg}$$

$$Q_r = m C_p (T_4 - T_1) = 1.005 (820.906 - 293)$$

$$\boxed{Q_r = 530.097 kJ/kg}$$

$$\eta_{Brayton} = 1 - \frac{1}{r_p^{\left(\frac{\gamma-1}{\gamma}\right)}}$$

$$= 1 - \frac{1}{3^{\frac{1.4-1}{1.4}}}$$

$$\boxed{\begin{aligned}\eta_{Brayton} &= 0.269 \\ &= 26.9\%\end{aligned}}$$

$$\eta_{Brayton} = \frac{W}{Q_s}$$

$$W = \eta_{Brayton} \times Q_s$$

$$= 0.269 \times 725.239$$

$$\boxed{W = 195.089 \text{ kJ/kg.}}$$

5. **An engine working on Otto cycle has a volume 0.45 m^3, pressure I bar and temperature 30°C at the beginning of compression stroke. At the end of compression stroke, the pressure is 11 bar. 210 kJ of heat is added at constant volume. Determine;(a) Pressures, temperatures and volumes at salient points in the cycle; (b) Percentage clearance, (c) Efficiency, (d) Network per cycle, (e) Mean effective pressure and (f) Ideal power developed by the engine if the number of working cycles per minute is 210. Assume the cycle is reversible. (NOV 2016)**

Solution. Refer Fig. 13.12

Volume,	$V_1 = 0.45 \text{ m}^3$
Initial pressure,	$p_1 = 1$ bar
Initial temperature,	$T_1 = 30 + 273 = 303$ K
Pressure at the end of compression stroke,	$p_2 = 11$ bar
Heat added at constant volume	$= 210$ kJ
Number of working cycles/min.	$= 210.$

(*i*) **Pressures, temperatures and volumes at salient points :**

For *adiabatic compression 1-2*

$$p_1V_1^{\gamma} = p_2V_2^{\gamma}$$

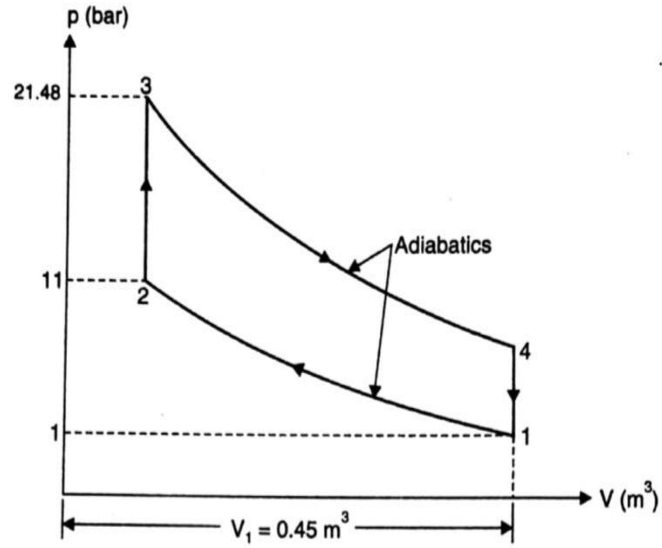


Fig. 13.12

or $$\frac{p_2}{p_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = (r)^{\gamma} \quad \text{or} \quad r = \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}} = \left(\frac{11}{1}\right)^{\frac{1}{1.4}} = (11)^{0.714} = 5.5$$

Also $$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = (r)^{\gamma-1} = (5.5)^{1.4-1} = 1.977 \simeq 1.98$$

$\therefore$ $$T_2 = T_1 \times 1.98 = 303 \times 1.98 = \mathbf{600\ K.} \quad \textbf{(Ans.)}$$

Applying gas laws to points 1 and 2

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

$\therefore$ $$V_2 = \frac{T_2}{T_1} \times \frac{p_1}{p_2} \times V_1 = \frac{600 \times 1 \times 0.45}{303 \times 11} = \mathbf{0.081\ m^3.} \quad \textbf{(Ans.)}$$

The heat supplied during the *process 2-3* is given by :

$$Q_s = m\,c_v\,(T_3 - T_2)$$

where $$m = \frac{p_1V_1}{RT_1} = \frac{1 \times 10^5 \times 0.45}{287 \times 303} = 0.517 \text{ kg}$$

$\therefore$ $$210 = 0.517 \times 0.71\,(T_3 - 600)$$

or $$T_3 = \frac{210}{0.517 \times 0.71} + 600 = \mathbf{1172\ K.} \quad \textbf{(Ans.)}$$

For the *constant volume process 2-3*

$$\frac{p_3}{T_3} = \frac{p_2}{T_2}$$

$\therefore$ $$p_3 = \frac{T_3}{T_2} \times p_2 = \frac{1172}{600} \times 11 = \mathbf{21.48\ bar.} \quad \textbf{(Ans.)}$$

$$V_3 = V_2 = \mathbf{0.081\ m^3.} \quad \textbf{(Ans.)}$$

For the *adiabatic (or isentropic) process 3-4*

$$p_3V_3^{\gamma} = p_4V_4^{\gamma}$$

$$p_4 = p_3 \times \left(\frac{V_3}{V_4}\right)^{\gamma} = p_3 \times \left(\frac{1}{r}\right)^{\gamma}$$

$$= 21.48 \times \left(\frac{1}{5.5}\right)^{1.4} = \mathbf{1.97\ bar.}\quad \textbf{(Ans.)}$$

Also
$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} = \left(\frac{1}{r}\right)^{\gamma-1} = \left(\frac{1}{5.5}\right)^{1.4-1} = 0.505$$

$\therefore$
$$T_4 = 0.505\ T_3 = 0.505 \times 1172 = \mathbf{591.8\ K.}\quad \textbf{(Ans.)}$$

$$V_4 = V_1 = \mathbf{0.45\ m^3.}\quad \textbf{(Ans.)}$$

(*ii*) **Percentage clearance :**

Percentage clearance

$$= \frac{V_c}{V_s} = \frac{V_2}{V_1 - V_2} \times 100 = \frac{0.081}{0.45 - 0.081} \times 100$$

$$= \mathbf{21.95\%.}\quad \textbf{(Ans.)}$$

(*iii*) **Efficiency :**

The heat rejected per cycle is given by

$$Q_r = mc_v(T_4 - T_1)$$

$$= 0.517 \times 0.71\ (591.8 - 303) = 106\ \text{kJ}$$

The air-standard efficiency of the cycle is given by

$$\eta_{otto} = \frac{Q_s - Q_r}{Q_s} = \frac{210 - 106}{210} = \mathbf{0.495\ or\ 49.5\%.}\quad \textbf{(Ans.)}$$

[Alternatively :

$$\eta_{otto} = 1 - \frac{1}{(r)^{\gamma-1}} = 1 - \frac{1}{(5.5)^{1.4-1}} = \mathbf{0.495\ or\ 49.5\%.}\quad \textbf{(Ans.)}$$]

(*iv*) **Mean effective pressure, $\mathbf{p_m}$:**

The mean effective pressure is given by

$$p_m = \frac{W\ \text{(work done)}}{V_s\text{(swept volume)}} = \frac{Q_s - Q_r}{(V_1 - V_2)}$$

$$= \frac{(210 - 106) \times 10^3}{(0.45 - 0.081) \times 10^5} = \mathbf{2.818\ bar.}\quad \textbf{(Ans.)}$$

(*v*) **Power developed, P :**

Power developed, P = Work done per second

= Work done per cycle × number of cycles per second

= (210 – 106) × (210/60) = **364 kW.** **(Ans.)**

6. **Derive the expression for an air standard efficiency and mean effective pressure of Dual cycle.(NOV 2016, MAY 2018, MAY 2019)**

This cycle (also called the *limited pressure cycle or mixed cycle*) is a combination of Otto and Diesel cycles, in a way, that heat is added partly at constant volume and partly at constant pressure ; *the advantage of which is that more time is available to fuel (which is injected into the engine cylinder before the end of compression stroke) for combustion. Because of lagging characteristics of fuel this cycle is invariably used for diesel and hot spot ignition engines.*

The dual combustion cycle (Fig. 13.19) consists of the following **operations** :

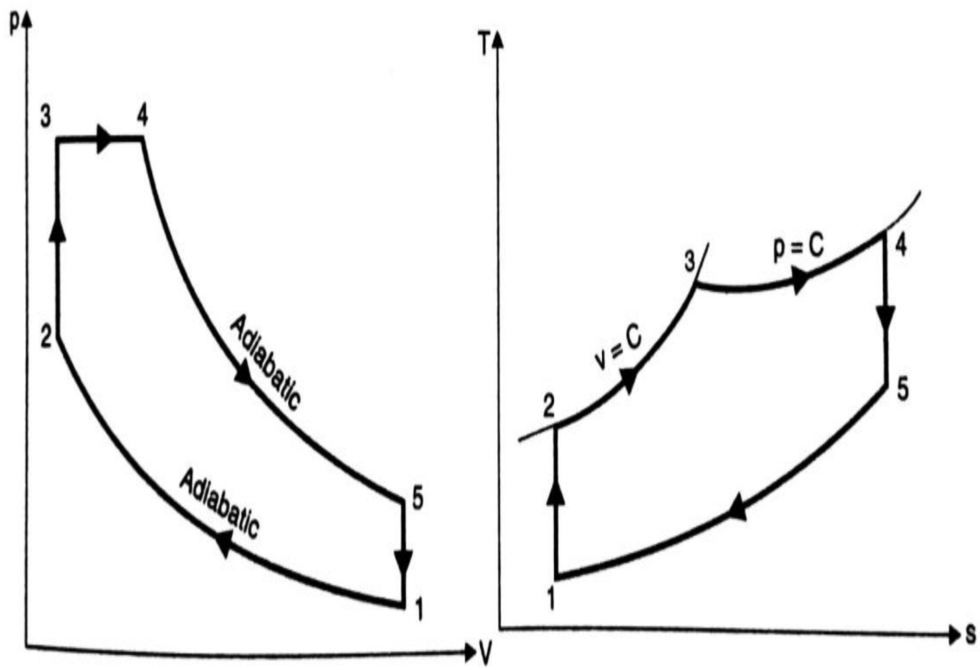
(*i*) 1-2—*Adiabatic compression*

(*ii*) 2-3—*Addition of heat at constant volume*

(*iii*) 3-4—*Addition of heat at constant pressure*

(*iv*) 4-5—*Adiabatic expansion*

(*v*) 5-1—*Rejection of heat at constant volume.*



Consider 1 kg of air.

Total heat supplied = Heat supplied during the operation 2-3 + heat supplied during the operation 3-4

$$= c_v(T_3 - T_2) + c_p(T_4 - T_3)$$

Heat rejected during operation 5-1 $= c_v(T_5 - T_1)$

Work done = Heat supplied − heat rejected

$$= c_v(T_3 - T_2) + c_p(T_4 - T_3) - c_v(T_5 - T_1)$$

$$\eta_{\text{dual}} = \frac{\text{Work done}}{\text{Heat supplied}} = \frac{c_v(T_3 - T_2) + c_p(T_4 - T_3) - c_v(T_5 - T_1)}{c_v(T_3 - T_2) + c_p(T_4 - T_3)}$$

$$= 1 - \frac{c_v(T_5 - T_1)}{c_v(T_3 - T_2) + c_p(T_4 - T_3)}$$

$$= 1 - \frac{c_v(T_5 - T_1)}{(T_3 - T_2) + \gamma(T_4 - T_3)} \quad \ldots(i) \quad \left(\because \ \gamma = \frac{c_p}{c_v}\right)$$

Compression ratio, $r = \dfrac{v_1}{v_2}$

During *adiabatic compression process 1-2,*

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma - 1} = (r)^{\gamma - 1} \quad \ldots(ii)$$

During *constant volume heating process,*

$$\frac{p_3}{T_3} = \frac{p_2}{T_2}$$

or $\dfrac{T_3}{T_2} = \dfrac{p_3}{p_2} = \beta$, where β is known as **pressure or explosion ratio.**

or

$$T_2 = \frac{T_3}{\beta} \quad \ldots(iii)$$

During *adiabatic expansion process,*

$$\frac{T_4}{T_5} = \left(\frac{v_5}{v_4}\right)^{\gamma - 1}$$

$$= \left(\frac{r}{\rho}\right)^{\gamma - 1} \quad \ldots(iv)$$

$$\left(\because \ \frac{v_5}{v_4} = \frac{v_1}{v_4} = \frac{v_1}{v_2} \times \frac{v_2}{v_4} = \frac{v_1}{v_2} \times \frac{v_3}{v_4} = \frac{r}{\rho}, \ \rho \text{ being the cut-off ratio}\right)$$

During *constant pressure heating process,*

$$\frac{v_3}{T_3} = \frac{v_4}{T_4}$$

$$T_4 = T_3 \frac{v_4}{v_3} = \rho\, T_3 \quad \ldots(v)$$

Putting the value of T_4 in the eqn. (*iv*), we get

$$\frac{\rho T_3}{T_5} = \left(\frac{r}{\rho}\right)^{\gamma - 1} \quad \text{or} \quad T_5 = \rho \cdot T_3 \cdot \left(\frac{\rho}{r}\right)^{\gamma - 1}$$

Putting the value of T_2 in eqn. (*ii*), we get

$$\frac{\frac{T_3}{\beta}}{T_1} = (r)^{\gamma - 1}$$

$$T_1 = \frac{T_3}{\beta} \cdot \frac{1}{(r)^{\gamma - 1}}$$

Now inserting the values of T_1, T_2, T_4 and T_5 in eqn. (*i*), we get

$$\eta_{dual} = 1 - \frac{\left[\rho \,.\, T_3 \left(\frac{\rho}{r}\right)^{\gamma-1} - \frac{T_3}{\beta} \cdot \frac{1}{(r)^{\gamma-1}}\right]}{\left[\left(T_3 - \frac{T_3}{\beta}\right) + \gamma(\rho T_3 - T_3)\right]} = 1 - \frac{\frac{1}{(r)^{\gamma-1}}\left(\rho^{\gamma} - \frac{1}{\beta}\right)}{\left(1 - \frac{1}{\beta}\right) + \gamma(\rho - 1)}$$

i.e.,

$$\eta_{dual} = 1 - \frac{1}{(r)^{\gamma-1}} \cdot \frac{(\beta \,.\, \rho^{\gamma} - 1)}{[(\beta - 1) + \beta\gamma(\rho - 1)]} \qquad ...(13.10)$$

Work done is given by,

$$W = p_3(v_4 - v_3) + \frac{p_4 v_4 - p_5 v_5}{\gamma - 1} - \frac{p_2 v_2 - p_1 v_1}{\gamma - 1}$$

$$= p_3 v_3(\rho - 1) + \frac{(p_4 \rho v_3 - p_5 r v_3) - (p_2 v_3 - p_1 r v_3)}{\gamma - 1}$$

$$= \frac{p_3 v_3 (\rho - 1)(\gamma - 1) + p_4 v_3 \left(\rho - \frac{p_5}{p_4} r\right) - p_2 v_3 \left(1 - \frac{p_1}{p_2} r\right)}{\gamma - 1}$$

Also $\quad \frac{p_5}{p_4} = \left(\frac{v_4}{v_5}\right)^{\gamma} = \left(\frac{\rho}{r}\right)^{\gamma}$ and $\frac{p_2}{p_1} = \left(\frac{v_1}{v_2}\right)^{\gamma} = r^{\gamma}$

also, $\quad p_3 = p_4$, $v_2 = v_3$, $v_5 = v_1$

$\therefore$

$$W = \frac{v_3[p_3(\rho - 1)(\gamma - 1) + p_3(\rho - \rho^{\gamma} r^{1-\gamma}) - p_2(1 - r^{1-\gamma})]}{(\gamma - 1)}$$

$$= \frac{p_2 v_2[\beta(\rho - 1)(\gamma - 1) + \beta(\rho - \rho^{\gamma} r^{1-\gamma}) - (1 - r^{1-\gamma})]}{(\gamma - 1)}$$

$$= \frac{p_1 (r)^{\gamma} v_1 / r[\beta\gamma(\rho - 1) + (\beta - 1) - r^{1-\gamma}(\beta\rho^{\gamma} - 1)]}{\gamma - 1}$$

$$= \frac{p_1 v_1 r^{\gamma - 1}[\beta\gamma(\rho - 1) + (\beta - 1) - r^{\gamma - 1}(\beta\rho^{\gamma} - 1)]}{\gamma - 1} \qquad ...(13.11)$$

Mean effective pressure (p_m) is given by,

$$p_m = \frac{W}{v_1 - v_2} = \frac{W}{v_1\left(\frac{r-1}{r}\right)} = \frac{p_1 v_1 [r^{1-\gamma}\beta\gamma(\rho - 1) + (\beta - 1) - r^{1-\gamma}(\beta\rho^{\gamma} - 1)]}{(\gamma - 1) v_1 \left(\frac{r-1}{r}\right)}$$

$$\mathbf{p_m} = \frac{p_1 (r)^{\gamma}[\beta(\rho - 1) + (\beta - 1) - r^{1-\gamma}(\beta\rho^{\gamma} - 1)]}{(\gamma - 1)(r - 1)} \qquad ...(13.12)$$

7. Derive the expression for an air standard efficiency and mean effective pressure of an otto cycle. (NOV 2016)

This cycle is so named as it was conceived by 'Otto'. On this cycle, petrol, gas and many types of oil engines work. It is the standard of comparison for internal combustion engines.

Figs. 13.5 (*a*) and (*b*) shows the theoretical *p-v* diagram and *T-s* diagrams of this cycle respectively.

- The point 1 represents that cylinder is full of air with volume V_1, pressure p_1 and absolute temperature T_1.
- Line 1-2 represents the *adiabatic compression* of air due to which p_1, V_1 and T_1 change to p_2, V_2 and T_2, respectively.
- Line 2-3 shows the *supply of heat* to the air *at constant volume* so that p_2 and T_2 change to p_3 and T_3 (V_3 being the same as V_2).
- Line 3-4 represents the *adiabatic expansion* of the air. During expansion p_3, V_3 and T_3 change to a final value of p_4, V_4 or V_1 and T_4, respectively.
- Line 4-1 shows the *rejection of heat* by air at *constant volume* till original state (point 1) reaches.

Consider **1 kg of air** (working substance) :

Heat supplied at constant volume $= c_v(T_3 - T_2)$.

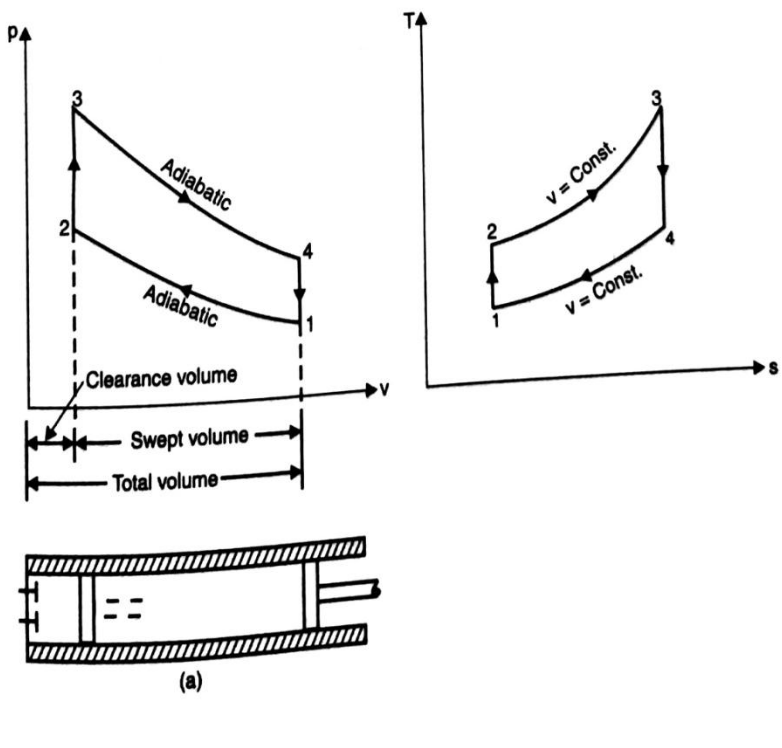
Heat rejected at constant volume $= c_v (T_4 - T_1)$.

But, work done = Heat supplied – Heat rejected

$$= c_v (T_3 - T_2) - c_v (T_4 - T_1)$$

$$\therefore \quad \text{Efficiency} = \frac{\text{Work done}}{\text{Heat supplied}} = \frac{c_v (T_3 - T_2) - c_v (T_4 - T_1)}{c_v (T_3 - T_2)}$$

$$= 1 - \frac{T_4 - T_1}{T_3 - T_2} \qquad ...(i)$$



(a)

Let compression ratio, $r_c\ (= r) = \dfrac{v_1}{v_2}$

and expansion ratio, $r_e\ (= r) = \dfrac{v_4}{v_3}$

(These *two ratios are same* in this cycle)

As $$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

Then, $$T_2 = T_1 \cdot (r)^{\gamma-1}$$

Similarly, $$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma-1}$$

or $$T_3 = T_4 \cdot (r)^{\gamma-1}$$

Inserting the values of T_2 and T_3 in equation (*i*), we get

$$\eta_{otto} = 1 - \frac{T_4 - T_1}{T_4 \cdot (r)^{\gamma-1} - T_1 \cdot (r)^{\gamma-1}} = 1 - \frac{T_4 - T_1}{r^{\gamma-1}(T_4 - T_1)}$$

$$= 1 - \frac{1}{(r)^{\gamma-1}} \quad \text{...(13.3)}$$

This expression is known as the ***air standard efficiency of the Otto cycle.***

It is clear from the above expression that efficiency increases with the increase in the value of r, which means we can have maximum efficiency by increasing r to a considerable extent, but *due to practical difficulties its value is limited to about 8.*

The *net work done per kg* in the Otto cycle can also be expressed in terms of p, v. If p is expressed in bar *i.e.,* 10^5 N/m², then work done

$$W = \left(\frac{p_3v_3 - p_4v_4}{\gamma-1} - \frac{p_2v_2 - p_1v_1}{\gamma-1}\right) \times 10^2 \text{ kJ} \quad \text{...(13.4)}$$

Also $$\frac{p_3}{p_4} = r^{\gamma} = \frac{p_2}{p_1}$$

$\therefore$ $$\frac{p_3}{p_2} = \frac{p_4}{p_1} = r_p$$

where r_p stands for *pressure ratio.* $\left[\because \dfrac{v_1}{v_2} = \dfrac{v_4}{v_3} = r\right]$

and $$v_1 = rv_2 = v_4 = rv_3$$

$\therefore$ $$W = \frac{1}{\gamma-1}\left[p_4v_4\left(\frac{p_3v_3}{p_4v_4} - 1\right) - p_1v_1\left(\frac{p_2v_2}{p_1v_1} - 1\right)\right]$$

$$= \frac{1}{\gamma-1}\left[p_4v_4\left(\frac{p_3}{p_4r} - 1\right) - p_1v_1\left(\frac{p_2}{p_1r} - 1\right)\right]$$

$$= \frac{v_1}{\gamma-1}\left[p_4\,(r^{\gamma-1} - 1) - p_1\,(r^{\gamma-1} - 1)\right]$$

$$= \frac{v_1}{\gamma - 1}\left[(r^{\gamma-1} - 1)(p_4 - p_1)\right]$$

$$= \frac{p_1 v_1}{\gamma - 1}\left[(r^{\gamma-1} - 1)(r_p - 1)\right] \qquad \text{...[13.4 (a)]}$$

Mean effective pressure (p_m) is given by :

$$p_m = \left[\left(\frac{p_3 v_3 - p_4 v_4}{\gamma - 1} - \frac{p_2 v_2 - p_1 v_1}{\gamma - 1}\right) \div (v_1 - v_2)\right] \text{ bar} \qquad \text{...(13.5)}$$

Also

$$p_m = \frac{\left[\frac{p_1 v_1}{\gamma - 1}(r^{\gamma-1} - 1)(r_p - 1)\right]}{(v_1 - v_2)}$$

$$= \frac{\frac{p_1 v_1}{\gamma - 1}[(r^{\gamma-1} - 1)(r_p - 1)}{v_1 - \frac{v_1}{r}}$$

$$= \frac{\frac{p_1 v_1}{\gamma - 1}[(r^{\gamma-1} - 1)(r_p - 1)]}{v_1\left(\frac{r-1}{r}\right)}$$

$$\mathbf{p_m} = \frac{p_1 r[(r^{\gamma-1} - 1)(r_p - 1)]}{(\gamma - 1)(r - 1)} \qquad \text{...(13.6)}$$

8. **In a diesel cycle the lowest temperature and pressure are 300 K and 1 bar respectively. The compression ratio is 16 and heat added per kg of air is 2500 kJ/kg. Determine,**
   a. **Temperature and pressure at the end of the compression,**
   b. **Maximum temperature reached in the cycle,**
   c. **Cycle efficiency,**
   d. **Power output for air flow rate of 0.25 kg/s. Take' C_P = 1 kJ/kgK and Cv= 0.714 kJ/kgK. (NOV 2018)**

given data

$P_1 = 1\text{ bar} = 100\text{ kn/m}^2$

$T_1 = 300K$

$r = V_1/V_2 = 16$

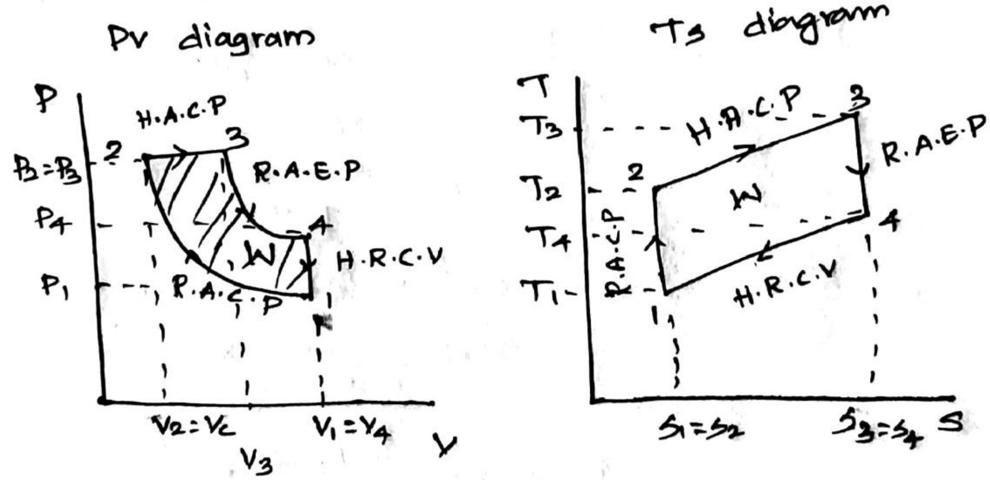
$Q_{2-3} = 2500\text{ kJ/kg}$

Diesel cycle.

To find

$T_2,\ P_2,\ T_3,\ \eta,\ P$

Sol.



1-2 Process Reversible Adiabatic Process.

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} \quad (T\ \&\ V\ \text{relation})$$

$$\gamma = C_P/C_V = 1/0.714$$

$$\gamma = 1.4$$

$$\frac{300}{T_2} = (1/16)^{1.4-1}$$

$$\boxed{T_2 = 911.854\text{ K}}$$

Pressure & Volume Relation.

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

$$\frac{100}{P_2} = \left(1/16\right)^{1.4}$$

$$\boxed{P_2 = 4854.368 \text{ kn/m}^2}$$

2-3 heat addition at Constant Pressure.

$$Q_{2-3} = m c_p (T_3 - T_2)$$

$$Q_{2-3} = 1 \times (T_3 - T_2)$$

$$2500 = 1 \times (T_3 - 911.854)$$

$$\boxed{T_3 = 3411.824 \text{ K}}$$

Temperature & volume relation for heat addition at constant pressure.

$$\frac{V_2}{T_2} = \frac{V_3}{T_3}$$

$$\frac{V_3}{V_2} = \frac{T_3}{T_2}.$$

$$V_3/V_2 = \rho$$

$$\rho = T_3/T_2 = \frac{3411.824}{911.854}$$

$$\rho = 3.741$$

$$\eta_{diesel} = 1 - \frac{1}{\gamma (r)^{\gamma-1}} \times \left[\frac{\rho^{\gamma} - 1}{\rho - 1}\right]$$

$$\eta_{diesel} = 1 - \frac{1}{1.4 (16)^{1.4-1}} \times \left[\frac{3.741^{1.4} - 1}{3.741 - 1}\right]$$

$$\boxed{\eta_{diesel} = 0.542 \quad 54\%}$$

$W = ?$

$$\eta_{diesel} = \frac{W}{Q_s}$$

$$W = Q_s \times \eta_{diesel}$$

$$W = 2500 \times 0.54$$

$$W = 1355$$

$$P = W \times m = 1355 \times 0.25$$

$$\boxed{P = 338.44 \text{ W}}$$

9. The minimum pressure and temperature in an Otto cycle are 100 kPa and 27 °C. The amount of heat added to the air per cycle is 1500 kJ/kg. Determine, (a) the pressure and temperature at all points of the air standard Otto cycle, (b) Specific work and (c) Thermal efficiency of the cycle for compression ratio of 8:1. Take for air, C_P = 0.72 kJ/kg K and γ = 1.4. (MAY 2018, MAY 2016)

Solution. Refer Fig. 13.7. *Given* : p_1 = 100 kPa = 10^5 N/m² or 1 bar ;
T_1 = 27 + 273 = 300 K ; Heat added = 1500 kJ/kg ;
r = 8 : 1 ; c_v = 0.72 kJ/kg ; γ = 1.4.

Consider 1 kg of air.

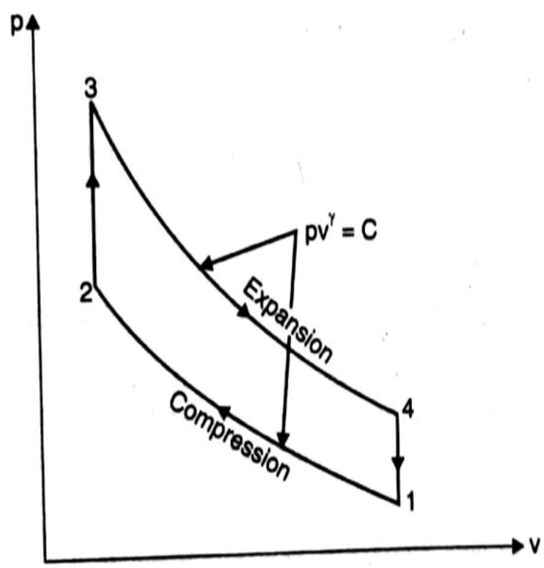


Fig. 13.7

(*i*) **Pressures and temperatures at all points :**

Adiabatic compression process 1-2 :

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma - 1} = (r)^{\gamma - 1} = (8)^{1.4 - 1} = 2.297$$

∴ $$\mathbf{T_2} = 300 \times 2.297 = \mathbf{689.1\ K.} \quad \textbf{(Ans.)}$$

Also $$p_1 v_1^{\gamma} = p_2 v_2^{\gamma}$$

or $$\frac{p_2}{p_1} = \left(\frac{v_1}{v_2}\right)^{\gamma} = (8)^{1.4} = 18.379$$

∴ $$\mathbf{p_2} = 1 \times 18.379 = \mathbf{18.379\ bar.} \quad \textbf{(Ans.)}$$

Constant volume process 2-3 :

Heat added during the process,

$$c_v (T_3 - T_2) = 1500$$

$$0.72\,(T_3 - 689.1) = 1500$$

$$\mathbf{T_3} = \frac{1500}{0.72} + 689.1 = \mathbf{2772.4\ K.} \quad \textbf{(Ans.)}$$

Also, $$\frac{p_2}{T_2} = \frac{p_3}{T_3} \Rightarrow \mathbf{p_3} = \frac{p_2 T_3}{T_2} = \frac{18.379 \times 2772.4}{689.1} = \mathbf{73.94\ bar.} \quad \textbf{(Ans.)}$$

Adiabatic Expansion process 3-4 :

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma - 1} = (r)^{\gamma - 1} = (8)^{1.4 - 1} = 2.297$$

$\therefore$ $$\mathbf{T_4} = \frac{T_3}{2.297} = \frac{2772.4}{2.297} = \mathbf{1206.9\ K.} \quad \textbf{(Ans.)}$$

Also, $$p_3 v_3^{\gamma} = p_4 v_4^{\gamma} \Rightarrow \mathbf{p_4} = p_3 \times \left(\frac{v_3}{v_4}\right)^{\gamma} = 73.94 \times \left(\frac{1}{8}\right)^{1.4} = \mathbf{4.023\ bar.} \quad \textbf{(Ans.)}$$

(*ii*) **Specific work and thermal efficiency :**

Specific work = Heat added − heat rejected

$$= c_v\,(T_3 - T_2) - c_v(T_4 - T_1) = c_v\,[(T_3 - T_2) - (T_4 - T_1)]$$
$$= 0.72\,[(2772.4 - 689.1) - (1206.9 - 300)] = \mathbf{847\ kJ/kg.} \quad \textbf{(Ans.)}$$

Thermal efficiency, $$\eta_{th} = 1 - \frac{1}{(r)^{\gamma - 1}}$$

$$= 1 - \frac{1}{(8)^{1.4-1}} = \mathbf{0.5647} \quad \textbf{or} \quad \mathbf{56.47\%.} \quad \textbf{(Ans.)}$$

10. An engine working on the Otto cycle is supplied with air at 0.1 MPa, 35°C. The compression ratio is 8.heat supplied is 2100 kJ/kg. Calculate the maximum pressure and temperature of the cycle, the cycle efficiency, and the mean effective pressure.(N0V 2018)

given data

$P_1 = 0.1\,mpa = 100\,kn/m^2$

$T_1 = 35^\circ c = 308\,K$

$r = \frac{V_1}{V_2} = 8$

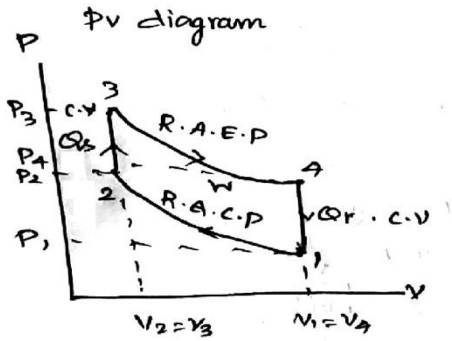
$Q_{2-3} = 2100\,kJ/kg$

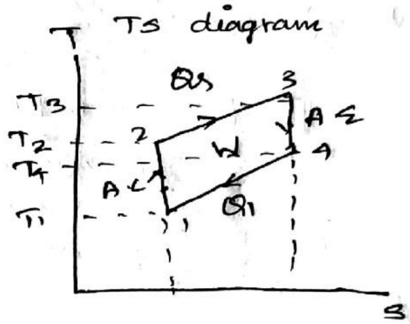
Otto cycle

To find

$P_3, T_3, \eta_{otto}, P_m = ?$

Sol





1-2 process Reversible Adiabatic process

T & V relation

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = (r)^{\gamma-1}$$

$$\frac{T_1}{T_2} = 1/8^{1.4-1}$$

$$T_2 = \frac{308}{1/8^{0.4}}$$

$$T_2 = 708.045\,K$$

P & V relation

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = r^{\gamma}$$

$$P_2 = 100 \times (8)^{1.4}$$

$$P_2 = 1837.9\,kn/m^2$$

2-3 process heat addition at constant volume

$$Q_{2-3} = mC_v(T_3 - T_2)$$

$$2100 = 1 \times 0.718(T_3 - T_2)$$

$$2100 = 0.718\,T_3 - 508.376$$

$$T_3 = 2608.376/1.005$$

$$\boxed{T_3 = 3632.835\ K}$$

P & T relation

$$P_2/P_3 = T_2/T_3$$

$$\frac{1837.9}{P_3} = \frac{508.376}{3632.835}$$

$$\boxed{P_3 = 13222.92\ kN/m^2}$$

$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$$

$$= 1 - \frac{1}{8^{1.4-1}}$$

$$\boxed{\eta_{otto} = 0.565 = 56.5\ \%}$$

Mean Effective pressure

$$P_m = \frac{\text{Workdone}}{\text{Swept Volume}} = \frac{W.D}{V_1 - V_2}.$$

W.D

$$\eta_{otto} = W.D/Q_s = W.D = \eta_{otto} \times Q_s$$

$$= 0.565 \times 2100$$

$$W = 1186.5\ kJ/kg$$

$$P_1V_1 = mRT_1$$

$$100 \times V_1 = 1 \times 0.287 \times 308$$

$$V_1 = 0.883\ m^3/kg$$

$$r = V_1/V_2 = 8$$

$$V_2 = \frac{0.883}{8}$$

$$V_2 = 0.110\ m^3/kg$$

$$P_m = \frac{1186.5}{0.883 - 0.110}$$

$$\boxed{P_m = 1534.92\ kN/m^2}$$

11. A dual combustion engines uses a compression ratio of 13.5, an explosion ratio of 1.5 and cutoff at 5 percent of the stroke. The bore and stroke of the cylinder are 20.5 and 41 cm respectively. Determine the ideal cycle efficiency.(MAY 2017)

given data

$r = \frac{V_1}{V_2} = 13.5$

explosion ratio $= \frac{P_3}{P_2} = 1.5 = r_p$

Cut off take 5 percent of the stroke.

$V_4 - V_3 = 5\% \; V_1 - V_2$

$d = 20.5$ cm $\quad 0.205$ m

$L = 41$ cm $\quad 0.41$ m

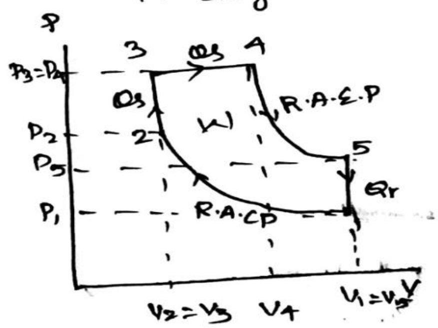
dual cycle

To find

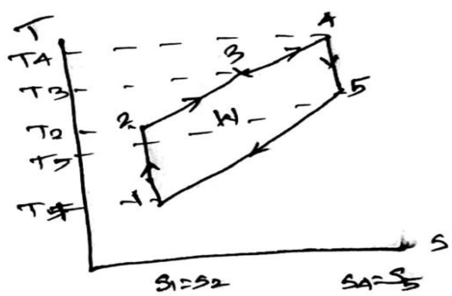
η_{dual}.

Sol

PV diagram



Ts diagram



Cut off ratio.

cut off takes place 5 % of stroke

$V_4 - V_3 = 0.05\,(V_1 - V_2)$

$V_4 - V_2 = 0.05\; V_1 - V_2$

$\boxed{V_2 = V_3 \text{ constant volume process}}$

$V_1/V_2 = 13.5$

$V_1 = 13.5 V_2$

$V_4 - V_2 = 0.05\,(13.5 V_2 - V_2)$

$$V_4 - V_2 = 0.05\,(12.5\,V_2)$$

$$V_4 - V_2 = 0.625\,V_2$$

$$V_4 = 0.625\,V_2 + V_2$$

$$V_4 = 1.625\,V_2$$

$$\rho = V_4/V_2 = V_4/V_3 = 1.625 \qquad V_2 = V_3.$$

Expansion ratio r/ρ

$$= \frac{13.5}{1.625}$$

$$r/\rho = 8.307 = V_5/V_4$$

$$\eta_{dual} = 1 - \left[\frac{1 \times (r_p \times \rho^{\gamma} - 1)}{r^{\gamma-1}\left[r_p - 1 + \gamma \times r_p(\rho - 1)\right]}\right]$$

$$= 1 - \left[\frac{1 \times (1.5 \times 1.625^{1.4} - 1)}{13.5^{1.4-1}\left[1.5 - 1 + 1.4 \times 1.5\,[1.625 - 1]\right]}\right]$$

$$\boxed{\eta_{dual} = 0.77 = 77\%}$$

12. Consider an air standard Otto cycle that has a heat addition of 2000 kJ/kg of air. Its compression ratio is 8. The pressure and temperature of air at the beginning of compression are 1 bar and 25°C. Determine for the cycle: (a) Maximum pressure and temperature (b) Thermal Efficiency (c) Mean effective pressure. (MAY 2017)

given data

$Q_{s\,2-3} = 2000\ kJ/kg$

$r = V_1/V_2 = 8$

$P_1 = 1\,bar = 100\ kN/m^2$

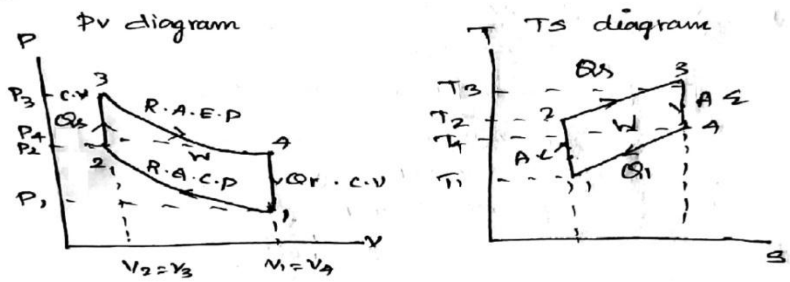
$T_1 = 25°C = 298\ K$

To find

Otto cycle

$P_3, T_3, \eta_{otto}, P_m = ?$

Sol



1-2 Process Reversible Adiabatic process

T & V relation

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = r^{\gamma-1}$$

$$T_2 = 298 \times (8)^{1.4-1}$$

$$T_2 = 684.62\ K$$

P & V relation

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = (r)^{\gamma}$$

$$P_2 = 100 \times 8^{1.4}$$

$$P_2 = 1837.917\ kN/m^2$$

2-3 Process heat addition at constant volume process

$$Q_{s\,2-3} = m\,C_v\,(T_3 - T_2)$$

$$2000 = 0.718\,(T_3 - 684.62)$$

$$\boxed{T_3 = 3509.41\ K}$$

T & P relation

$$\frac{P_2}{P_3} = \frac{T_2}{T_3}$$

$$\frac{1837.917}{P_3} = \frac{684.62}{3509.41}$$

$$\boxed{P_3 = 9425.215 \text{ Kn/m}^2}$$

$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$$

$$= 1 - \frac{1}{8^{1.4-1}}$$

$$\boxed{\eta_{otto} = 0.565 = 56.5\%}$$

Mean effective pressure P_m

$$P_m = \frac{\text{Workdone}}{\text{Swept Volume}} = \frac{W.D}{V_1 - V_2}$$

$$\eta_{otto} = \frac{W.D}{Q_S}$$

$$W.D = \eta_{otto} \times Q_S$$

$$W.D = 0.565 \times 2000$$

$$W.D = 1130 \text{ KJ/kg}$$

$V_1 = ?$

$$P_1 V_1 = mRT_1$$

$$100 \times V_1 = 1 \times 0.287 \times 298$$

$$V_1 = 0.855 \text{ m}^3/\text{kg}$$

$$\frac{V_1}{V_2} = 8$$

$$V_2 = V_1/8 = \frac{0.855}{8}$$

$$V_2 = 0.106 \text{ m}^3/\text{kg}$$

$$P_m = \frac{1130}{0.855 - 0.106}$$

$$\boxed{P_m = 1508.678 \text{ Kn/m}^2}$$

13. A gas turbine plant operates on the Brayton cycle between T_{mln} = 33 K and T_{max} = 1073 K. Find the maximum work done per kg of air, and the corresponding cycle efficiency. How does this efficiency compare with the Carnot cycle efficiency operating between the same two temperatures? (NOV 2017)

Given data:-

$T_{min} = 300\,K$

$T_{max} = 1073\,K$

To find

$W_{max} = ?$

$\eta_{Brayton} = ?$

ratio of Brayton & Carnot efficiency

Sol.

$$W_{max} = c_p\left(\sqrt{T_{max}} - \sqrt{T_{min}}\right)^2$$

$$= 1.005\left(\sqrt{1073} - \sqrt{300}\right)^2$$

$$\boxed{W_{max} = 239.28\ KJ/kg}$$

$$\eta_{Brayton} = 1 - \sqrt{\frac{T_{min}}{T_{max}}}$$

$$= 1 - \sqrt{\frac{300}{1073}}$$

$$\boxed{\eta_{Brayton} = 0.47 \quad 47\%}$$

$$\eta_{carnot} = 1 - \frac{T_{min}}{T_{max}}$$

$$= 1 - \frac{300}{1073}$$

$$\eta_{carnot} = 0.721 = 72\%$$

$$\frac{\eta_{Brayton}}{\eta_{carnot}} = \frac{0.47}{0.721}$$

$$\boxed{\frac{\eta_{Brayton}}{\eta_{carnot}} = 0.652}$$

14. An engine works on Otto-cycle. The initial pressure and temperature of the air are 1 bar and 27^{0}C. Heat supplied at the end of compression is 850 kJ of heat per kg of air. Estimate the temperature and pressures at all salient points if the compression ratio is 6. Also find the efficiency of the cycle. Assume C_p = 1.005 kJ/kg K, C_v= 0.71 kJ/kg K and γ =1.4.(SEP 2020)

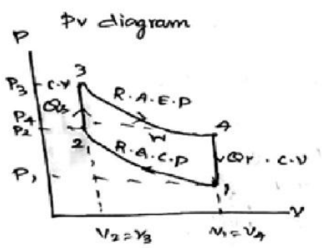
given data

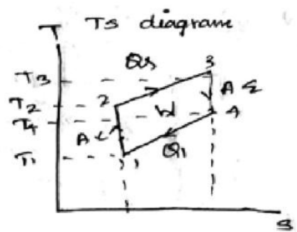
P_1 = 1 bar = 100 kPa	Otto cycle
T_1 = 27°C = 300 K	c_p = 1.005 kJ/kg K
Q_s = 850 kJ/kg	c_v = 0.71 kJ/kg K
$r = V_1/V_2 = 6$	γ = 1.4

To find

$T_2, T_3, P_2, P_3, \eta_{otto}$

sol





1-2 Reversible Adiabatic compression

Temperature & Volume relation

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$T_2 = T_1 \left(V_1/V_2\right)^{\gamma-1} \qquad V_1/V_2 = r = 6$$

$$= 300 \times (6)^{1.4-1}$$

$$\boxed{T_2 = 614.301\text{ K}}$$

Pressure & Volume relation

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$$

$$P_2 = P_1 \left(V_1/V_2\right)^{\gamma}$$

$$= 100 \times (6)^{0.4} \quad 1.4$$

$$\boxed{P_2 = 1228.603\text{ kN/m}^2}$$

2-3 process heat addition at constant volume

$$Q_s = m c_v (T_3 - T_2)$$

$$850 = 1 \times 0.71\,(T_3 - 614.301)$$

$$\boxed{T_3 = 1834.73\ K}$$

Pressure & Volume relation for 2-3 process

$$\frac{P_2}{P_3} = \frac{T_2}{T_3}$$

$$\frac{1228.603}{P_3} = \frac{614.301}{1834.73}$$

$$\boxed{P_3 = 3678.452\ Kn/m^2}$$

3-4 Process Reversible Adiabatic Expansion process.

Temperature & volume relation

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} \qquad \frac{V_4}{V_3} = r_c = 6$$

$$T_4 = T_3\,(V_3/V_4)^{\gamma-1}$$

$$= 1834.73 \times (1/6)^{1.4-1}$$

$$\boxed{T_4 = 821.959\ K}$$

Pressure & volume relation

$$P_4/P_3 = (V_3/V_4)^{\gamma}$$

$$P_4 = P_3\,(V_3/V_4)^{\gamma}$$

$$= 3678 \times (1/6)^{1.4}$$

$$\boxed{P_4 = 299.364\ Kn/m^2}$$

$$\eta_{otto} = 1 - \frac{1}{r_c^{\gamma-1}}$$

$$= 1 - \frac{1}{6^{1.4-1}}$$

$$\boxed{\eta_{otto} = 0.512 \quad 51.2\%}$$

15. In an air-standard Diesel cycle, the conditions at the start of the compression stroke are $v_1 = 0.05m^3$, $p_1 = 100Kpa$ and $T_1 = 300K$. The pressure at the end of compression stroke is 4 MPa. The heat addition is 500 kJ/kg of air. Determine the compression ratio, the cut-off ratio, the work done per kg of air and the thermal efficiency of the cycle. Assume $C_p = 1.005$ kJ/kg K, $C_v= 0.71$ kJ/kg K.(SEP 2020)

Given data

$V_1 = 0.05 m^3$

$P_1 = 100 kpa$

$T_1 = 300 K$

$P_2 = 4 MPa$ 400 kpa

$Q_S = 500 KJ/kg$

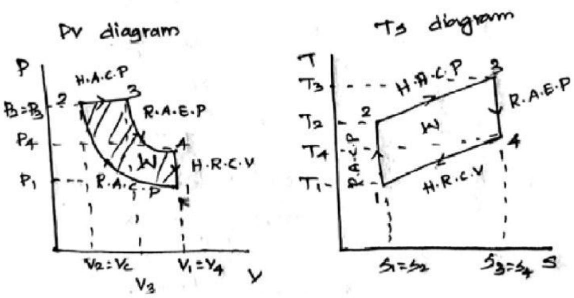
$Cp = 1.005 KJ/kg K$

$Cv = 0.41 KJ/kg K$

To find

$r, \rho, W, \eta_{diesel}$.

Sol



1-2 process Reversible Adiabatic process

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = \left(\frac{P_2}{P_1}\right)^{1/\gamma} = \frac{V_1}{V_2}$$

$$\left(\frac{4000}{100}\right)^{1/1.4} = \frac{V_1}{V_2}$$

$$\boxed{\frac{V_1}{V_2} = 13.941 = r}$$

Temperature & Volume relation

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$T_2 = T_1 (r)^{\gamma-1} \quad 0.4$$

$$= 300 \times 13.941$$

$$\boxed{T_2 = 860.674 K}$$

2-3 process Heat addition at constant pressure process.

$Q_s = mc_p (T_3 - T_2)$

$500 = 1 \times 1.005 (T_3 - 860.674)$

$\underline{T_3 = 1358.186 K}$

characteristic gas equation

$P_1 V_1 = mRT_1$

$100 \times V_1 = 1 \times 0.287 \times 300$

$\underline{V_1 = 0.861} \; m^3/kg$

$\frac{V_1}{V_2} = 13.941$

$\frac{0.861}{13.941} = V_2$

$\underline{V_2 = 0.0617} \; m^3/kg$

2-3 process

temperature & volume relation

$\frac{V_2}{V_3} = \frac{T_2}{T_3}$

$\frac{0.0617}{V_3} = \frac{860.674}{1358.186}$

$\underline{V_3 = 0.0979} \; m^3/kg$

$\frac{V_3}{V_2} = \rho = \frac{0.0979}{0.0617}$

$$\boxed{\rho = 1.58}$$

$$\eta_{diesel} = 1 - \frac{1}{\gamma}\left[\frac{\rho^{\gamma} - 1}{r^{\gamma-1} \times (\rho - 1)}\right]$$

$$= 1 - \frac{1}{1.4 (13.94)^{1.4-1}} \left[\frac{1.58^{1.4} - 1}{1.58 - 1}\right]$$

$$\boxed{\eta_{diesel} = 65.7\% \text{ (or) } 0.657}$$

$\eta_{diesel} = \frac{W.D}{Q_s}$

$0.657 = \frac{W.D}{500}$

$$\boxed{W.D = 328.5 \; kJ/kg}$$

16. **Air enters the compressor of a gas turbine plant operating on Brayton cycle at 101.325 kPa, 27°C. The pressure ratio in the cycle is 6. Calculate the maximum temperature in the cycle and the cycle efficiency. Assume $W_T = 2.5\ W_C$, where W_T and W_C are the turbine and the compressor work respectively. Take $\gamma = 1.4$.(NOV 2017)**

given data

$P_1 = 101.325$ kpa

$T_1 = 27°c$

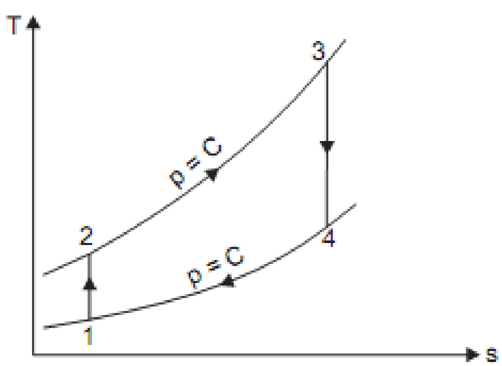
$P_2/P_1 = 6 = r_p$

$W_T = 2.5 W_c$ — W_T = Turbine work

$\gamma = 1.4$ — W_c = compressorwork

To Find

T_3, $\eta_{Brayton}$.



$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = (r_p)^{\frac{\gamma-1}{\gamma}} = (6)^{\frac{1.4-1}{1.4}} = 1.668$$

$$T_2 = 1.668\ T_1 = 1.668 \times 300 = 500.4\ \text{K}$$

Also, $$\frac{T_3}{T_4} = (r_p)^{\frac{\gamma - 1}{\gamma}} = (6)^{\frac{1.4-1}{1.4}} = 1.668$$

$\therefore$ $$T_4 = \frac{T_3}{1.668}$$

But, $$W_T = 2.5\ W_C$$

$\therefore$ $$mc_p\,(T_3 - T_4) = 2.5\ mc_p\,(T_2 - T_1)$$

$$T_3 - \frac{T_3}{1.668} = 2.5\,(500.4 - 300) = 501 \quad \text{or} \quad T_3\left(1 - \frac{1}{1.668}\right) = 501$$

$\therefore$ $$T_3 = \frac{501}{\left(1 - \frac{1}{1.668}\right)} = \textbf{1251 K or 978°C.} \quad \textbf{(Ans.)}$$

(*ii*) **Cycle efficiency, η_{cycle} :**

Now, $$T_4 = \frac{T_3}{1.668} = \frac{1251}{1.668} = 750\text{ K}$$

$$\eta_{cycle} = \frac{\text{Net work}}{\text{Heat added}} = \frac{mc_p\,(T_3 - T_4) - mc_p\,(T_2 - T_1)}{mc_p(T_3 - T_2)}$$

$$= \frac{(1251 - 750) - (500.4 - 300)}{(1251 - 500.4)} = \textbf{0.4 or 40\%.} \quad \textbf{(Ans.)}$$

$$\left[\ \textit{Check};\ \eta_{cycle} = 1 - \frac{1}{(r_p)^{\left(\frac{\gamma-1}{\gamma}\right)}} = 1 - \frac{1}{(6)^{\frac{1.4-1}{1.4}}} = \textbf{0.4 or 40\%.} \quad \textbf{(Ans.)}\ \right]$$